

Article

The Discovery of MVT-like Ga-Enriched Sphalerite from the Zhaojinci Area in the South Hunan District (South China)

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Abstract

Gallium (Ga) enrichment in sphalerite has been widely recognized; however, its enrichment mechanisms remain insufficiently understood. The South Hunan district, located at the intersection of the Nanling Region and the Qin-Hang Metallogenic Belt in South China, is characterized by abundant Jurassic magmatic-hydrothermal Pb–Zn deposits, which typically host Ga-depleted sphalerite. Recently, Ga-enriched sphalerite (up to 385 ppm by LA-ICP-MS) has been identified in newly drilled cores at Zhaojinci, adding complexity to the regional Pb–Zn metallogenic framework. EPMA elemental mapping and LA-ICP-MS time-resolved spectra indicate that Ga is homogeneously distributed within sphalerite, excluding the presence of micron-scale Ga-bearing mineral inclusions. A strong positive correlation between Ga and Cu concentrations suggests that Ga incorporation is facilitated by the coupled substitution of Zn^{2+} by Cu^+ . Sphalerite geothermometry yields formation temperatures of 118–138 °C (average 126 °C for GGIMF and ~129 °C for SPRFT), accompanied by intermediate sulfur fugacity conditions ($\lg f\text{S}_2 = -22.9$ to -21.2), which appear to favor Ga enrichment in sphalerite. The trace element geochemistry of the Zhaojinci sphalerite (Ga–Ge–Cd-enriched and Mn–In–Sn–Co-depleted), combined with its formation under low-temperature (120–180 °C) and intermediate $f\text{S}_2$ conditions (within the pyrite stability field), is consistent with MVT-like mineralization. This interpretation is supported by multiple lines of geological evidence, including the strict confinement of stratabound Pb–Zn mineralization to the Devonian Xikuangshan Formation limestone, structural control by syn-sedimentary normal faults, pervasive dolomitization of the host rocks, and the absence of genetic relationship to magmatic activity. Moreover, the sphalerite geochemical signature, corroborated by an XGBoost-based machine learning classifier, reinforces the MVT-like affinity for the Zhaojinci mineralization. This study not only emphasizes the importance of low-temperature and intermediate- $f\text{S}_2$ conditions in Ga enrichment within sphalerite, but also highlights the significance of discovering MVT-like sphalerite for Pb–Zn resource exploration in the South Hunan district, providing valuable new insights and directions for mineral prospecting in this geologically important region of South China.



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Keywords: Gallium; physicochemical conditions; MVT-like sphalerite; South Hunan district; South China

1. Introduction

Gallium (Ga) is a strategically critical metal widely used in semiconductor, photovoltaic, and other advanced technologies owing to its unique electronic and thermal properties [1–4]. In geological systems, Ga mainly occurs as a trace element in aluminosilicate minerals and sulfide phases, with sphalerite being one of its most important hosts [5–9]. The enrichment of Ga in sphalerite is generally defined by three criteria: (1) global Mississippi Valley-type (MVT) deposit standards (>100 ppm Ga), (2) regional MVT benchmarks (e.g., 100–300 ppm in South China), and (3) deposit-scale comparisons, where magmatic-hydrothermal sphalerite typically contains <50 ppm Ga [4–9]. Understanding the mechanisms of Ga enrichment in sphalerite is essential for evaluating its economic potential and for guiding exploration strategies [10–14].

Gallium enrichment in sphalerite is governed by multiple physicochemical factors, including temperature, redox state, fluid composition, and crystal-chemical substitution mechanisms [15–19]. Gallium is mainly transported as chloride complexes [20], and its enrichment is influenced by temperature, fluid mixing, and redox conditions. Elevated Cl concentrations and reducing environments favor Ga substitution in sphalerite [21], while moderate-to-low temperatures (<250 °C) promote Ga enrichment [6]. Exceptional Ga content has been reported in deposits such as Djebel Gustar (2108 ppm Ga) [21] and Shalipayco (3943 ppm Ga) [6].

The Nanling Region and Qin-Hang Belt in South China are renowned metallogenic provinces characterized by a diverse array of hydrothermal polymetallic deposits (Figure 1) [22–24]. The South Hunan district (Figure 2), strategically situated at the intersection of the Nanling Region and Qin-Hang Belt, represents a significant ore cluster, featuring numerous Jurassic magmatic-hydrothermal Pb–Zn deposits [25–28]. However, Ga concentrations in these deposits are notably low, as exemplified by the Baoshan (2.77–33.4 ppm) [29] and Dafang (7.20–15.37 ppm) [30] deposits, consistent with the generally low Ga abundances reported for magmatic-hydrothermal systems [31]. This understanding is now challenged by the recent discovery of Ga-enriched sphalerite in the Zhaojinci district, located between the Dafang Pb–Zn–Ag–Au and Baoshan Cu–Pb–Zn deposits (Figure 2). To resolve this inconsistency, this study integrates textural observations, EPMA major-element mapping, and LA-ICP-MS trace-element analyses of Ga-enriched sphalerite from Zhaojinci, with the aims of determining the occurrence and distribution of Ga, elucidating its enrichment mechanisms, constraining the physicochemical controls, and providing new insights into Pb–Zn metallogeny in the South Hunan district, South China.

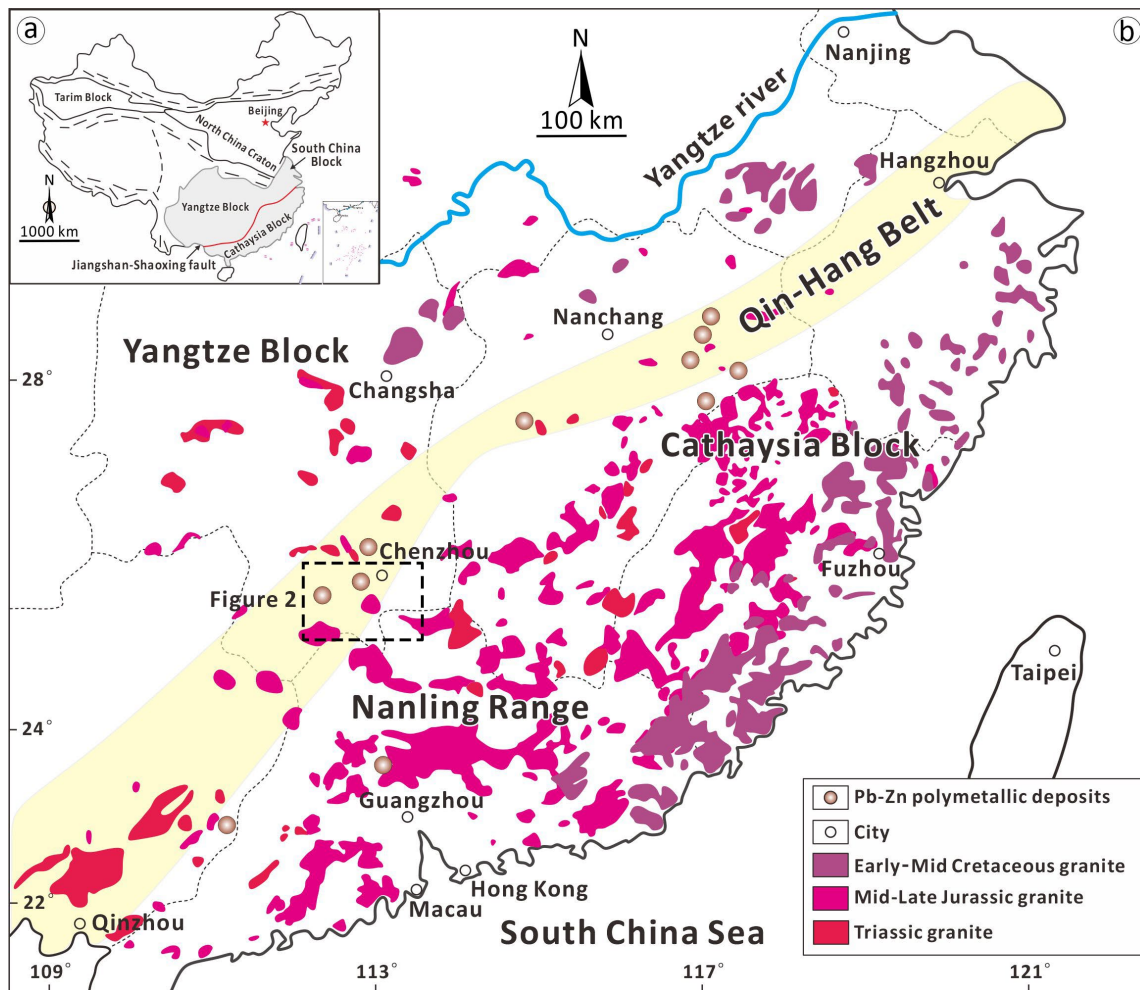


Figure 1. (a) Geologic map showing the location of South China in the Chinese mainland; (b) Locations of the Nanling Range and Qin-Hang Belt in South China [23,32].

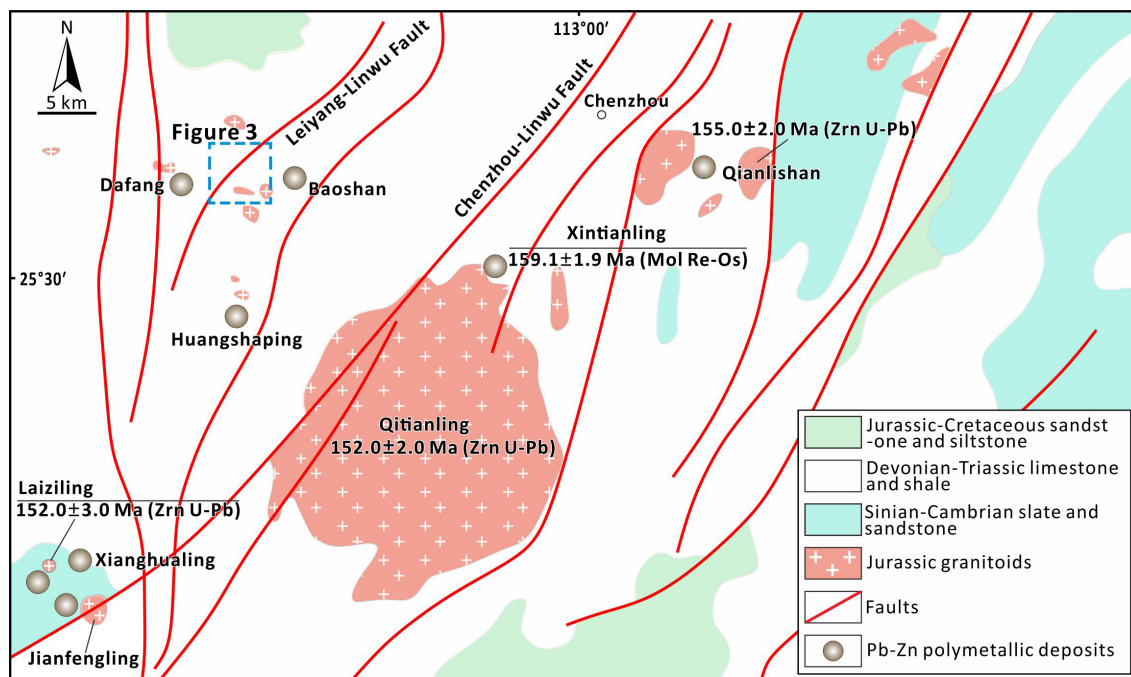


Figure 2. Geologic map of the South Hunan district [26]. The dotted-line quadrilateral marks the Zhaojinci exploration area (Figure 3). Mineral abbreviations: Zrn = zircon; Mol = molybdenite.

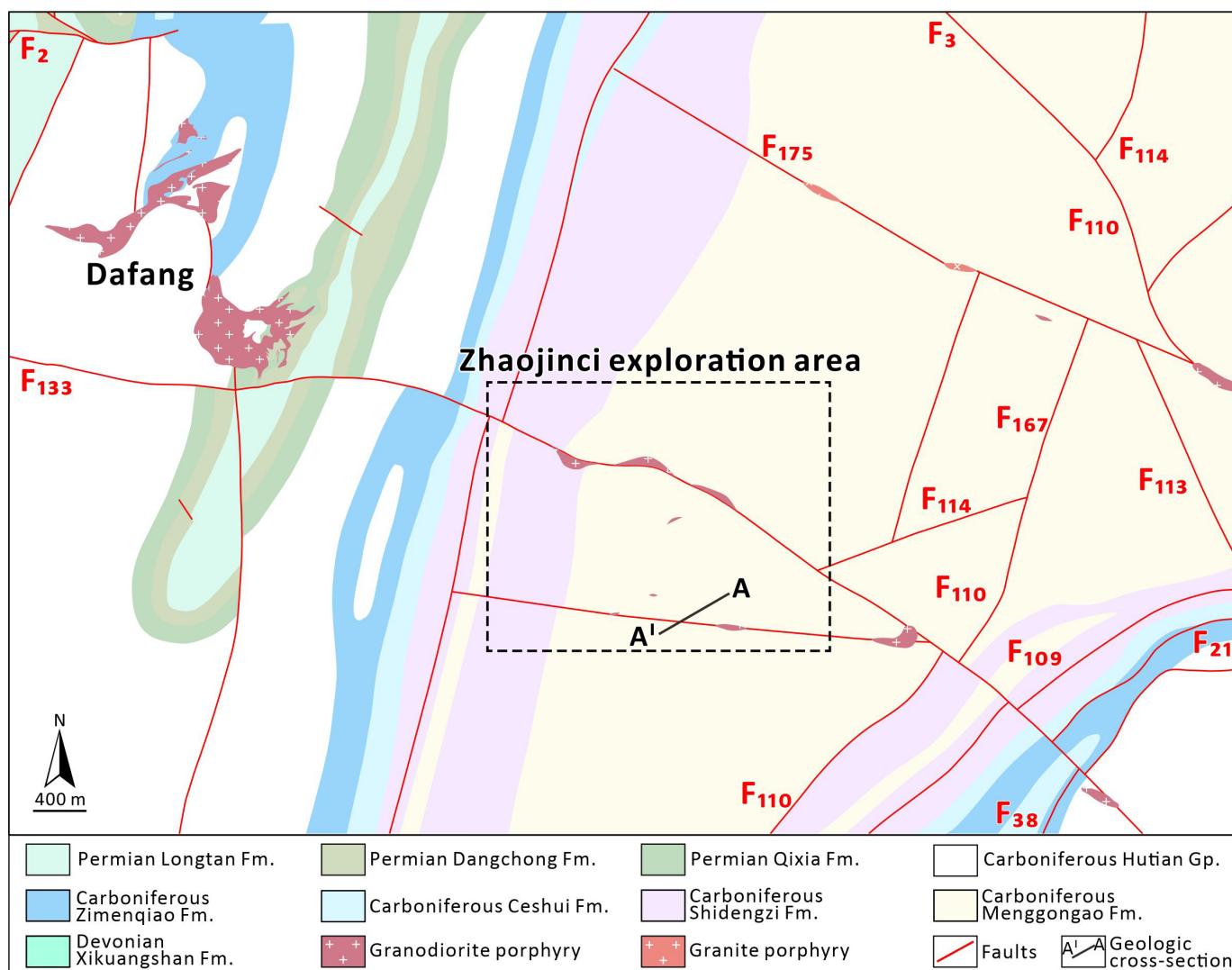


Figure 3. Geologic map of the Zhaojinci district [30,33].

2. Geological Setting

2.1. Regional Geology

The South China Block (SCB) is tectonically bounded by several prominent orogenic belts and suture zones. To the northwest, it is separated from the Songpan-Ganzi Terrane by the Longmenshan Fault. To the north, the Qinling–Dabie–Sulu Orogen marks its boundary with the North China Craton, while the Ailaoshan–Songma Suture Zone defines its southwestern margin, separating it from the Indochina Terrane (Figure 1a) [34]. The SCB is an amalgamation of the Yangtze Block (to the northwest) and the Cathaysia Block (to the southeast), resulting from a Neoproterozoic accretionary event demarcated by the Jiangshan–Shaoxing Fault (Figure 1a) [35,36]. Throughout its geological history, the SCB has experienced significant Mesozoic magmatism, which led to the widespread development of granitic rocks. This magmatic activity occurred in distinct episodes: the Triassic (ca. 251–205 Ma), Mid-Late Jurassic (ca. 180–142 Ma), and Cretaceous (ca. 140–66 Ma) (Figure 1b) [32]. These magmatic pulses exhibit both temporal and spatial associations with three distinct metallogenic epochs [37,38]. The initial mineralization event is constrained to the Late Triassic (230–210 Ma), followed by a subsequent phase during the Mid-Late Jurassic (170–150 Ma), and culminating with a final episode in the Early–Mid Cretaceous (120–80 Ma). Notably, the Mid-Late Jurassic tectonic-magmatic event was particularly

significant, giving rise to numerous porphyry-skarn Cu deposits along the Qin-Hang Belt, with formation ages typically ranging from 170 to 160 Ma. Concurrently, it was responsible for the extensive formation of granite-related W–Sn–Pb–Zn polymetallic deposits within the Nanling Region, primarily dated between 160 and 150 Ma (Figure 1b) [23,39,40].

The South Hunan district is strategically situated at the intersection of the Nanling Region and Qin-Hang Belt, representing a significant ore cluster, featuring abundant well-developed hydrothermal deposits (e.g., Huangshaping, Tongshanling, Dafang, Baoshan) [29,30,33,41,42]. The outcropping lithostratigraphic units in the South Hunan district predominantly consist of a sedimentary succession, including Devonian–Triassic limestone and shale (primary ore host), Triassic–Neogene sandstone and siltstone, and Sinian–Cambrian slate and sandstone (Figure 2) [26]. The geological evolution of the region has been shaped by multiple tectonic episodes and associated magmatism. The regional basement complex, primarily consisting of severely metamorphosed pre-Silurian sedimentary rocks, formed during and after the Caledonian orogeny (Late Silurian–Devonian), which resulted in dominant EW-trending folds and EW-/NE-trending fault systems. These basement rocks are unconformably overlain by a thick sequence of Devonian–Permian carbonate platforms. Following the Caledonian effects, the Indosinian orogeny (Triassic) led to the formation of a series of NS-trending folds. Subsequently, the Yanshanian orogeny (Jurassic–Cretaceous), driven by the rollback of the Paleo-Pacific subduction zone, further influenced regional tectonics, resulting in the development of NNE-trending basin rifts and associated fault networks [25,37,43].

In the South Hunan district, the Yanshanian tectonism was accompanied by extensive Jurassic granitic intrusions. These granitoids, including granite and quartz porphyry, manifest in various intrusive geometries such as batholiths, laccoliths, stocks, and dikes (Figure 2) [26]. The W–Sn polymetallic mineralization shows a genetic association with highly fractionated S-type granites (159–155 Ma), which are petrologically characterized by reduced oxygen fugacity (f_{O_2}), low water content, seagull-shaped rare earth element (REE) patterns, and magmatic evolution dominated by fractional crystallization processes [33]. Representative examples include the Qianlishan pluton (157.0 ± 2.0 Ma) [43] and Qitianling batholith (158.0 ± 1.1 Ma) [44]. In contrast, Cu–polymetallic mineralization is genetically linked to I-type granodioritic intrusions (162–160 Ma) that are distinguished by hydrous, oxidized magmatic signatures, listric-shaped REE patterns, and magmatic evolution governed primarily by partial melting processes [28]. The Baoshan stock (161.0 ± 1.3 Ma) [45] and Tongshanling porphyry complex (161.4 ± 0.9 Ma;) [46] serve as typical examples of this mineralization style.

2.2. The Zhaojinci Area

The Zhaojinci area is situated within the Pingbao orefield in the South Hunan district (Figure 2) [30,33,47,48], and positioned between the Dafang Pb–Zn–Au–Ag deposit (38 kt Pb @ 1.4%, 33 kt Zn @ 1.3%, 6.1 t Au @ 1.80 g/t, and 370 t Ag @ 106.8 g/t) [47,49] and Baoshan Cu–Pb–Zn deposit (23.9 kt Cu @ 1.28%, 47.1 kt Pb @ 3.82%, and 51.3 kt Zn @ 4.34%) [50,51]. This strategic location suggests the area holds significant potential for Pb–Zn mineral exploration. The lithostratigraphic sequence exposed in the region predominantly consists of Devonian to Permian sedimentary formations (Figure 3), including the Devonian Xikuangshan Formation (composed of shale and limestone), Menggongao Formation (marlstone and limestone), Shidengzi Formation (limestone), Ceshui Formation (sandstone), Zimenqiao Formation (dolostone), and the Carboniferous Hutianqun Formation (dolostone). The sequence continues with the Permian Qixia Formation (limestone), Dangchong Formation (shale and mudstone), and Longtan Formation (sandstone), repre-

senting a continuous depositional sequence from the Middle Devonian through the Late Permian [28,33,52].

The structural framework of the study area is dominated by a NE-trending tectonic pattern, with a series of anticlinal and synclinal structures, intersected by NE- and NW-trending fault systems (Figure 3) [46,53]. The fold architecture in the Zhaojinci region is principally defined by four major anticlines and one syncline: the Shangshuitou-Jiangjiabao, Weijia-Shangbao, Putang-Shanmuchong, and Dongshuitang anticlines, and the Dafang syncline [46]. Structural analysis identifies two distinct fault systems based on their spatial orientation: NE-trending and NW-trending faults. The NE-trending fault system, representing the primary tectonic features in the region, predates the NW-trending faults and includes significant structures such as F₆, F₁₁₆, F₁₁₄, F₁₁₀, F₁₆₇, F₁₁₃, F₁₀₉, F₂₁, and F₃₈. In contrast, the NW-trending fault system consists of smaller normal faults, primarily functioning as secondary structures that have controlled magmatic emplacement, with notable examples including F₁₃₃ and F₃ (Figure 3). This complex tectonic evolution, with multiple phases of deformation, has played a crucial role in the localization of mineralization processes in the region [50–52].

The Zhaojinci area hosts significant granitic intrusions, primarily comprising granite porphyry and granodiorite porphyry (Figure 3). The granite porphyry occurs as small-scale dikes that intrude into Carboniferous sedimentary sequences, with emplacement structurally controlled by the NW-trending secondary fault F₁₇₅. In contrast, the granodiorite porphyry is represented by 10 NW-trending plutons, which are concentrated in the central sector of the area. These plutons are spatially associated with fault structures F₁₃₃ and F₃, collectively forming the Dongshui intrusion (Figure 3). The structural control of these intrusions suggests a genetic relationship between faulting and magmatic activity, with the NW-trending faults serving as preferred pathways for magma ascent and emplacement [30,54,55].

3. Sampling and Analytical Methods

3.1. Samples

A comprehensive suite of 21 samples was systematically collected from various drill cores (QZB802, QZB801, QZB804, and ZKB801; Figure 4) within the Zhaojinci area. Petrographic analysis was performed on 13 polished thin sections, which were examined using both transmitted and reflected light microscopy. Mineralogical investigations identified sphalerite (50–200 µm in size), galena (10–30 µm in size), and pyrite (50–500 µm in size) assemblages within Devonian Xikuangshan Formation limestone (Figures 4 and 5a–c). These sulfides exhibit spatial association with dolomitization and textural relationships indicative of a later-stage hydrothermal event. Eight representative sphalerite-bearing samples were selected for further analysis.

In situ major element analyses were conducted on sphalerite from samples ZKB801-5, ZKB801-6, and ZKB801-7. Specifically, samples ZKB801-6 and ZKB801-7 were subjected to EPMA mapping to delineate elemental distribution patterns. Additionally, trace element compositions of sphalerite were also determined through in situ analyses of samples ZKB801-5, ZKB801-6, and ZKB801-7. Comprehensive sample descriptions, including lithological characteristics and mineralogical associations, are presented in Table 1.

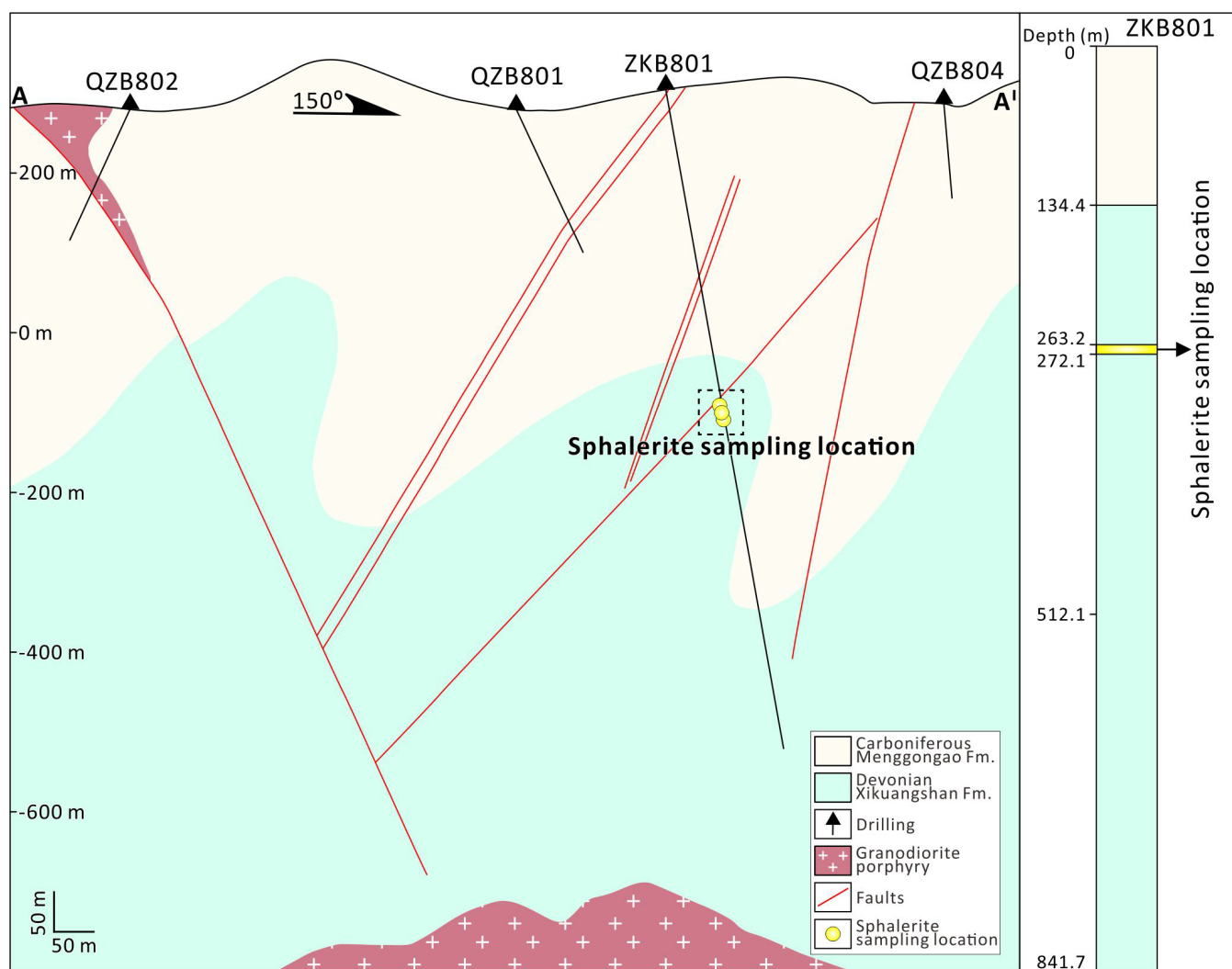


Figure 4. Geological cross-section of the Zhaojinci area and columnar section of drill hole ZKB801.

Table 1. Descriptions of representative sulfide samples from drill hole ZKB801 at Zhaojinci. Abbreviations: Py = pyrite; Gn = galena; Sp = sphalerite; Qz = quartz.

Sample No.	Mineral Assemblages	Description
ZKB801-5	Qz-Py-Gn-Sp	Euhedral-subhedral pyrite intergrown with sphalerite and galena
ZKB801-6	Qz-Py-Gn-Sp	Pyrite with some galena and sphalerite
ZKB801-7	Qz-Py-Gn-Sp	Pyrite intergrown with sphalerite and galena
ZKB801-9	Qz-Sp	Quartz-bearing massive sphalerite ore
ZKB801-10	Qz-Sp	Some sphalerite intergrown with quartz
ZKB801-11	Qz-Py-Sp	Quartz-bearing massive pyrite and sphalerite
ZKB801-12	Qz-Py-Sp	Euhedral-subhedral sphalerite intergrown with pyrite
ZKB801-13	Qz-Py-Sp	Some pyrite intergrown with sphalerite

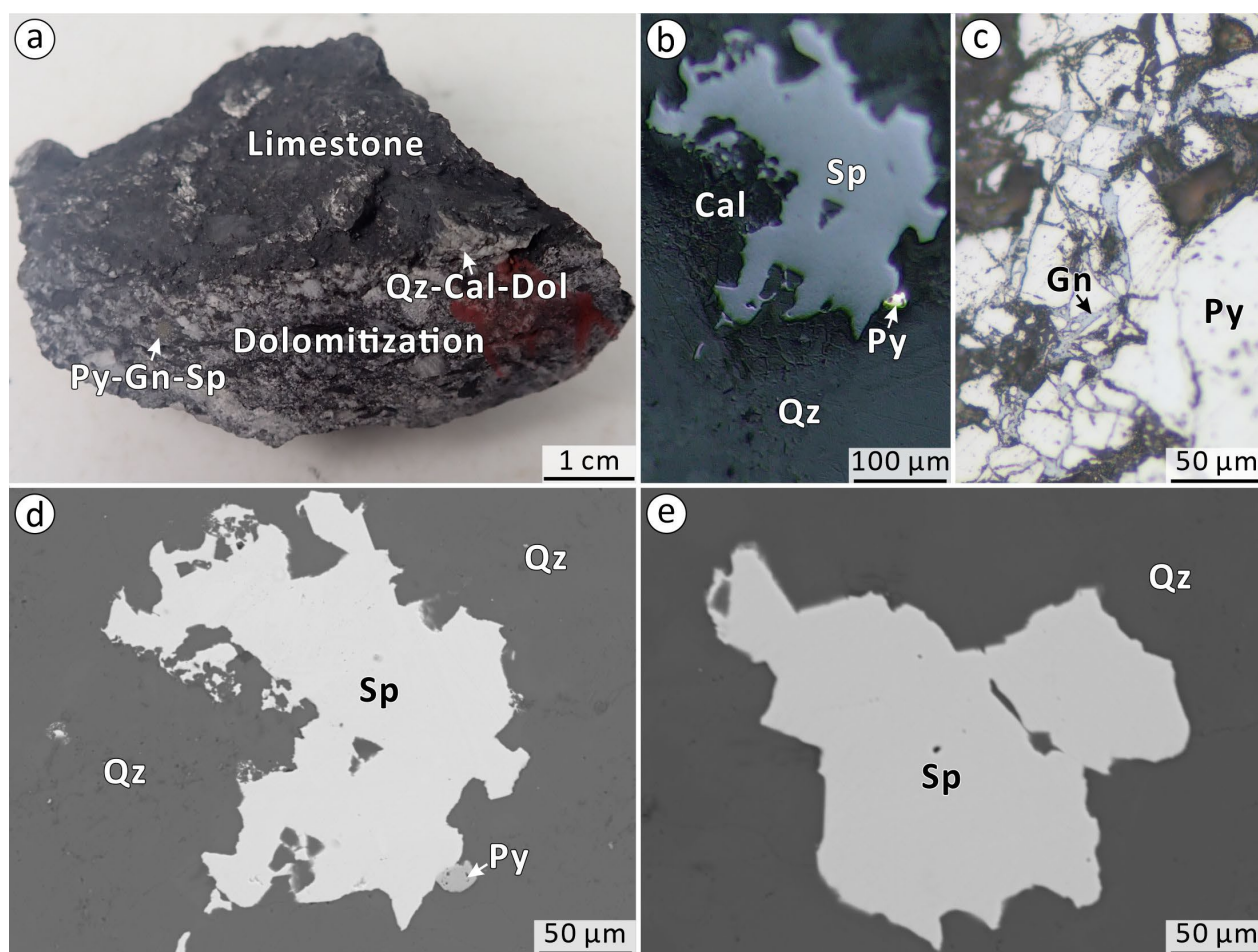


Figure 5. Representative alteration and mineralization images at Zhaojinci. (a) Hand sample photograph of the sphalerite-galena-pyrite paragenesis developed in zones of pervasive dolomitization; (b) Reflect-lighted image of sphalerite intergrown with minor pyrite; (c) Reflect-lighted image of galena occurs as fine veinlets crosscutting pyrite; (d) Backscattered electron images of sphalerite exhibiting a homogeneous internal texture (no zoning or replacement) and coexisting with minor pyrite; (e) Backscattered electron image of texturally homogenous sphalerite. Abbreviations: Py = pyrite; Gn = galena; Sp = sphalerite; Qz = quartz.

3.2. SEM-BSE Imaging

Polished thin sections were meticulously prepared for all samples before LA-ICP-MS analyses. The internal microstructure of sphalerite was investigated using two scanning electron microscopes (SEMs): a JCM-7000 SEM (JEOL Ltd., Tokyo, Japan) at Central South University (operating at 15 kV accelerating voltage and 10 nA beam current) and a TESCAN MIRA3 field emission-scanning electron microscope (FE-SEM) (TESCAN, Brno, Czech Republic) at Guangzhou Tuoyan Analytical Technology Co., Ltd. (operating at 20 kV accelerating voltage and 15 nA beam current). Backscattered electron (BSE) imaging facilitated detailed mineralogical characterization and textural analysis.

3.3. EPMA Spot Analysis and Elemental Mapping of Sphalerite

The major and trace element compositions of sphalerite were analyzed using a JEOL JXA-8230 electron probe microanalyzer (EPMA; JEOL Ltd., Tokyo, Japan) at the School of Geoscience and Technology, Southwest Petroleum University, Chengdu, China. Before analysis, the polished samples were coated with a uniform 20 nm-thick carbon film to ensure conductivity. The EPMA was operated under optimized analytical conditions, including an accelerating voltage of 20 kV, a beam current of 20 nA, and a focused beam

diameter of 1–2 μm . The analyzed elements included S, Zn, Fe, Cd, Mn, Pb, Ga, Ge, Cr, As, Co, Ni, Cu, Ag, In, Sn, and Sb, with calibration conducted using certified reference materials: FeS_2 (for S, Fe), ZnS (for Zn), CuFeS_2 (for Cu), pure Cd metal (for Cd), Mn metal (for Mn), PbS (for Pb), GaAs (for Ga), Ge metal (for Ge), Cr_2O_3 (for Cr), FeAsS (for As), Co metal (for Co), Ni metal (for Ni), Ag metal (for Ag), InAs (for In), SnO_2 (for Sn), and Sb_2S_3 (for Sb). All data were corrected using the ZAF (Z: atomic number, A: absorption, F: fluorescence) method. The estimated detection limits for trace elements by EPMA are approximately 0.01 wt.% for most elements analyzed. In addition, high-resolution X-ray elemental intensity maps for Zn, S, Fe, Ga, Cu, and Ag were acquired in wavelength-dispersive spectrometer (WDS) mode. The mapping parameters included an accelerating voltage of 20 kV, a beam current of 400 nA, a pixel size of 1 μm , and a dwell time of 100 ms per pixel, enabling detailed visualization of the elemental distribution within representative sphalerite grains.

3.4. LA-ICP-MS Trace Element Analyses of Sphalerite

In situ trace element analysis of sphalerite was conducted utilizing a NWR 193 nm ArF excimer laser ablation system (Teledyne CETAC Technologies, Omaha, NE, USA) coupled to an iCAP RQ inductively coupled plasma mass spectrometer (ICP-MS; Thermo Fisher Scientific, Bremen, Germany) at Guangzhou Tuoyan Analytical Technology Co., Ltd. Before analysis, all target spots were meticulously examined using BSE imaging to ensure exclusion of mineral inclusions and secondary phases. The ablation process employed helium as the carrier gas to optimize aerosol transport efficiency and minimize deposition within the ablation cell. The laser system was operated with optimized parameters, including a fluence of $3.5 \text{ J}/\text{cm}^2$, a repetition rate of 6 Hz, and a spot diameter of 30 μm . Each analysis comprised an 80 s time-resolved measurement sequence, incorporating 15 s for background signal acquisition, 40 s for active sample ablation, and 25 s for residual signal washout. A suite of 19 isotopes was quantified for sphalerite analysis: ^{34}S , ^{55}Mn , ^{57}Fe , ^{59}Co , ^{60}Ni , ^{65}Cu , ^{66}Zn , ^{71}Ga , ^{74}Ge , ^{77}Se , ^{97}Mo , ^{107}Ag , ^{111}Cd , ^{115}In , ^{118}Sn , ^{121}Sb , ^{126}Te , ^{208}Pb , and ^{209}Bi . The Iolite software package (Iolite 4) [56] was employed for data reduction, implementing user-defined time intervals for baseline correction to derive session-wide baseline-corrected values for each measured isotope. The analytical sequence followed a standardized protocol, consisting of two reference materials (NIST 610 and GSE-2G) and one sulfide standard (MASS-1), followed by batches of 5 to 8 unknown samples. Zinc concentrations determined by EMPA were used as internal standards for quantitative normalization.

4. Results

4.1. Internal Texture of Sphalerite

Using BSE imaging, sphalerite from the Zhaojinci prospect appears light gray and exhibits a homogeneous internal texture, without observable zoning or replacement textures (Figure 5d,e). It is commonly intergrown with fine-grained pyrite in a close coexisting association (Figure 5d).

4.2. Major Element Geochemistry of Sphalerite

A total of 10 EMPA spot analyses were conducted on the Zhaojinci sphalerite, with the complete major-element compositional data listed in Table 2. The sphalerite exhibits a relatively consistent chemical composition, characterized by Zn (64.38–65.67 wt.%), S (32.30–33.21 wt.%), and Cd (0.24–0.29 wt.%) concentrations. Notably, the sphalerite is commonly enriched in Ga (up to 0.18 wt.%) and Fe (0.90–1.99 wt.%). Concentrations of Ge, In, Sn, and Sb were found to be near or below the detection limits of the EPMA. Furthermore, EMPA elemental mapping reveals homogeneous distributions of Zn, S, Fe,

Ga, Ge, and Cu within the sphalerite grains (Figure 6), with no discernible anomalies or zoning patterns observed.

Table 2. Major-element data obtained by EPMA for the Zhaojinci sphalerite.

Samples	Points	S wt. %	Zn wt. %	Ga wt. %	Fe wt. %	Cu wt. %	Ge wt. %	Ag wt. %	Cd wt. %	In wt. %	Sn wt. %	Sb wt. %	Total wt. %
ZKB801-5	1	32.30	65.28	0.10	1.22	<LOD	<LOD	<LOD	0.28	<LOD	<LOD	0.01	99.18
ZKB801-5	2	32.41	65.38	0.03	1.57	0.03	<LOD	0.02	0.29	<LOD	0.01	0.01	99.74
ZKB801-5	3	33.12	65.50	0.07	1.07	0.06	<LOD	<LOD	0.24	<LOD	0.01	<LOD	100.07
ZKB801-6	4	32.48	65.67	0.05	1.04	0.06	<LOD	<LOD	0.26	<LOD	<LOD	<LOD	99.57
ZKB801-6	5	32.68	65.24	0.18	1.23	0.08	<LOD	<LOD	0.27	0.01	0.01	<LOD	99.69
ZKB801-6	6	32.50	64.79	0.08	1.17	0.09	<LOD	0.01	0.25	<LOD	0.01	<LOD	98.91
ZKB801-7	7	33.21	64.89	0.10	0.90	0.07	0.02	<LOD	0.26	<LOD	0.01	0.02	99.47
ZKB801-7	8	32.59	64.84	0.09	1.99	<LOD	<LOD	<LOD	0.24	<LOD	0.01	<LOD	99.76
ZKB801-7	9	32.41	64.84	0.12	1.28	<LOD	<LOD	0.02	0.27	<LOD	0.02	<LOD	98.95
ZKB801-7	10	32.69	64.38	<LOD	1.98	0.04	<LOD	<LOD	0.25	<LOD	<LOD	0.01	99.35
Mean values		32.64	65.08	0.09	1.35	0.06	0.02	0.02	0.26	0.01	0.01	0.01	99.47

The estimated detection limits (LOD, in wt. %) for the analyzed elements are as follows: S (0.01), Zn (0.01), Ga (0.01), Fe (0.01), Cu (0.01), Ge (0.01), Ag (0.01), Cd (0.01), In (0.01), Sn (0.01), Sb (0.01).

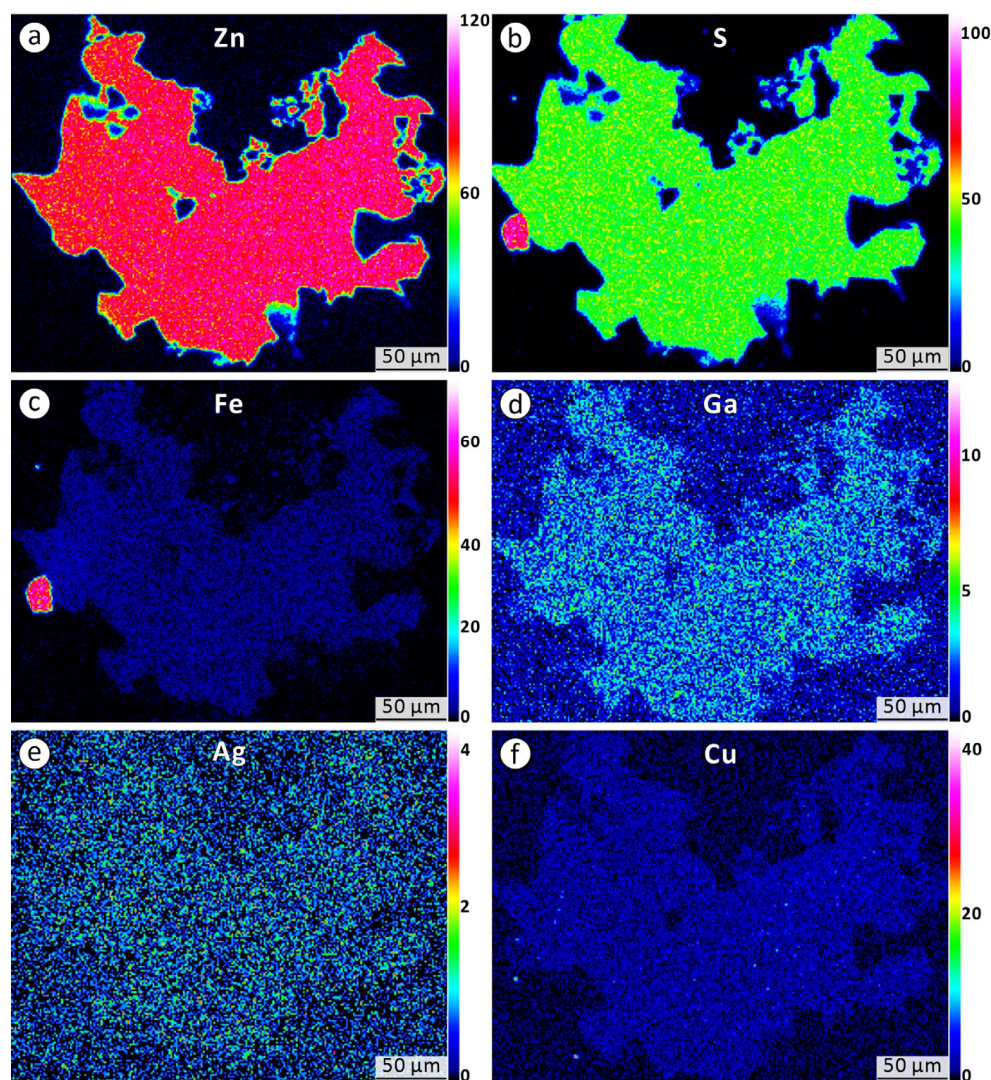


Figure 6. EPMA elemental mapping of sphalerite showing the homogeneous distribution of Zn (a), S (b), Fe (c), Ga (d), Ag (e), and Cu (f). Areas in (b,c) correspond to pyrite (Py); the corresponding BSE image is shown in Figure 5d.

4.3. Trace Element Geochemistry of Sphalerite

Fourteen LA-ICP-MS spot analyses were conducted on the Zhaojinci sphalerite samples, with the detailed trace-element compositions documented in Table 3 and graphically represented in Figure 7. Sphalerite exhibits elevated median concentrations of Fe (2395 ppm), Cd (2376 ppm), Cu (396 ppm), and Ga (126 ppm). Notably, it shows a pronounced depletion in Pb (38.7 ppm), Ag (0.956 ppm), and Sb (0.137 ppm). Other elements demonstrate relatively low values (Se: 1.05 ppm; Ni: 0.576 ppm; Co: 0.563 ppm; Bi: 0.033 ppm; Mo: 0.009 ppm), while Ge (30.3 ppm) and In (27.1 ppm) are present at moderate levels, all compared to typical magmatic-hydrothermal sphalerite [4–9].

Table 3. Trace-element data obtained by LA-ICP-MS for sphalerite from Zhaojinci.

Samples	Points	Mn ppm	Fe ppm	Co ppm	Ni ppm	Cu ppm	Ga ppm	Ge ppm	Se ppm	Mo ppm	Ag ppm	Cd ppm	In ppm	Sn ppm	Sb ppm	Pb ppm	Bi ppm
ZKB801-5	1	13.1	2209	0.544	0.241	1308	303	227	0.357	0.018	2.15	2589	45.7	14.8	0.516	142	0.256
ZKB801-5	2	6.07	2353	0.412	0.586	1017	385	199	1.78	b.d.l.	1.62	2519	41.9	7.38	0.891	117	0.221
ZKB801-5	3	6.34	2412	0.451	0.441	530	116	57.0	1.24	b.d.l.	1.33	2502	33.6	1.81	0.206	79.7	0.123
ZKB801-5	4	17.0	2224	0.441	0.411	691	223	101	0.409	b.d.l.	1.78	2457	36.7	8.75	0.387	90.0	0.176
ZKB801-6	5	6.67	2386	0.545	0.523	385	98.8	33.7	1.64	0.007	1.06	2436	28.2	1.89	0.115	45.5	0.071
ZKB801-6	6	7.53	2472	0.607	0.623	295	80.0	20.4	0.637	b.d.l.	0.95	2427	27.2	1.23	0.058	24.0	0.025
ZKB801-6	7	8.98	2388	0.581	0.422	407	126	24.7	0.858	b.d.l.	0.936	2352	27.0	1.82	0.079	29.1	0.030
ZKB801-6	8	5.95	2870	1.49	0.732	266	116	34.5	1.61	b.d.l.	0.845	2400	16.1	8.25	b.d.l.	41.7	0.012
ZKB801-6	9	14.7	2146	0.447	0.380	284	66.4	20.4	1.0	b.d.l.	1.07	2347	29.4	4.34	0.160	46.4	0.058
ZKB801-7	10	15.7	2383	0.599	0.604	267	126	23.0	0.583	0.012	0.892	2334	26.0	3.09	0.068	21.0	0.034
ZKB801-7	11	17.4	2402	0.526	0.566	319	119	26.9	1.05	b.d.l.	0.961	2269	26.8	4.89	0.067	25.3	0.032
ZKB801-7	12	5.81	3216	2.88	0.843	221	136	25.0	b.d.l.	0.011	0.785	2330	8.96	7.53	b.d.l.	31.5	0.012
ZKB801-7	13	9.12	3318	3.35	0.632	1506	198	21.0	2.28	0.443	0.887	2290	9.89	16.6	0.441	35.6	0.015
ZKB801-7	14	18.7	3138	2.34	1.00	695	187	38.6	2.50	0.004	0.852	2312	14.8	5.01	0.002	32.0	0.012
Median values		9.05	2395	0.563	0.576	396	126	30.3	1.05	0.011	0.956	2376	27.1	4.95	0.137	38.7	0.033

b.d.l. = Below detection limit.

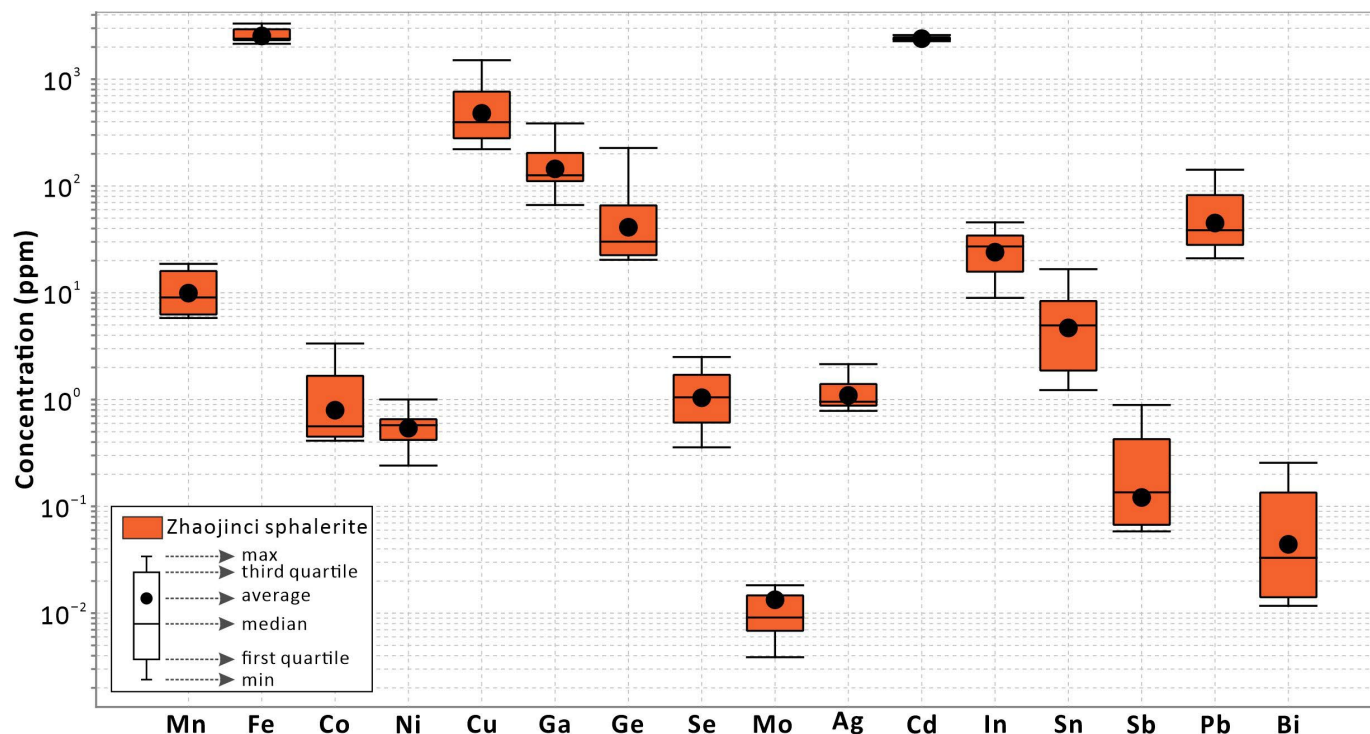


Figure 7. Box-whisker diagram illustrating the trace-element composition of Zhaojinci sphalerite.

The Fe content measured by EPMA (0.90–1.99 wt.%) is systematically higher than the median value obtained by LA-ICP-MS (2395 ppm). This discrepancy is best explained by the analytical characteristics of the two techniques. EPMA, while excellent for major

elements, has higher detection limits and is less accurate for trace-level concentrations compared to LA-ICP-MS. Therefore, the LA-ICP-MS data are considered more reliable for quantitative trace element analysis, and all subsequent discussions and comparisons are based exclusively on these results. Similar discrepancies are observed for Ga, Ge, Ag, Sn, and Sb. For instance, the maximum EPMA Ga (0.18 wt.%) is significantly higher than the LA-ICP-MS median (126 ppm), reflecting EPMA's higher detection limits and reduced accuracy for trace elements. Therefore, LA-ICP-MS data are preferred for trace element analysis, and all comparative studies rely exclusively on LA-ICP-MS results.

5. Discussion

5.1. Gallium Occurrence in the Zhaojinci Sphalerite

Sphalerite is commonly documented to contain micro-/nano-inclusions, as evidenced by numerous studies [57–60]. The presence of such inclusions can significantly obscure the intrinsic geochemical signatures of the host mineral [61–63], highlighting the need for a thorough evaluation of their impact on the overall geochemical composition of sphalerite. LA-ICP-MS time-resolved signal spectra have proven effective for identifying the occurrence and spatial distribution patterns of trace elements within sphalerite [64,65]. In the case of Zhaojinci sphalerite, the time-resolved signal spectra of Ga exhibit consistently flat and stable profiles, suggesting the absence of Ga-bearing micro-inclusions and indicating that Ga is uniformly distributed within the sphalerite lattice through solid solution. This finding aligns well with the EPMA mapping results, which demonstrate homogeneous Ga distribution throughout the sphalerite matrix without evidence of micron-scale inclusions (Figure 6d). The concordance between these two independent analytical techniques provides robust evidence for the lattice-bound incorporation of Ga in the Zhaojinci sphalerite, rather than its occurrence in discrete mineral phases.

Previous research has elucidated that the incorporation of Ga^{3+} into the sphalerite lattice occurs primarily through coupled substitution mechanisms, involving replacement of Zn^{2+} with various species such as Ag^+ , Cu^+ , In^{3+} , and structural vacancies [66–70]. Geochemical analyses reveal that Ga^{3+} concentration does not exhibit significant correlations with trivalent In (Figure 8a) nor monovalent Ag (Figure 8b). However, a distinct positive correlation is observed between Ga and Cu^+ ($R^2 = 0.63$, $p < 0.05$; Figure 8c,d), with the majority of data points plotting above the $\text{Cu}/\text{Ga} = 1:1$ stoichiometric line (Figure 8c). The positive correlation between Ga and Cu concentrations suggests that Ga^{3+} incorporates into the sphalerite lattice via a coupled substitution mechanism with Cu^+ , effectively replacing two Zn^{2+} ions [16,18,71].

5.2. Enrichment Mechanisms of Gallium in Sphalerite

Gallium enrichment in sphalerite is influenced by a range of factors, including fluid composition, crystallization kinetics, and physicochemical conditions (temperature, $f\text{O}_2$, $f\text{S}_2$) [31,72–75]. At Zhaojinci, the absence of micro- or nano-inclusions in sphalerite suggests that rapid crystallization was not the primary process for Ga enrichment [14,76]. Similarly, the pronounced depletion of Pb-Ag-Sb argues against fluid mixing as a dominant mechanism, as Cl-complexed elements such as Pb, Ag, and Sb would typically be enriched in such scenarios [21,59,77–80].

Instead, the enrichment of Ga is closely linked to the specific physicochemical conditions under which the Zhaojinci sphalerite formed. Geothermometric calculations based on sphalerite composition (GGIMFis and SPRFT methods) indicate low crystallization temperatures of 118–138 °C [81–83], consistent with those reported for typical MVT deposits [84,85]. Furthermore, sulfur fugacity estimates derived from Fe content and temperature constraints place the mineralization within the intermediate-sulfidation range ($\lg f\text{S}_2 = -22.9$

to -21.2) [86–88]. The combination of low temperature and intermediate fS_2 appears to have favored the incorporation of Ga into the sphalerite lattice, a pattern also recognized in several other Ga-rich MVT-like systems [6,73,84,85].

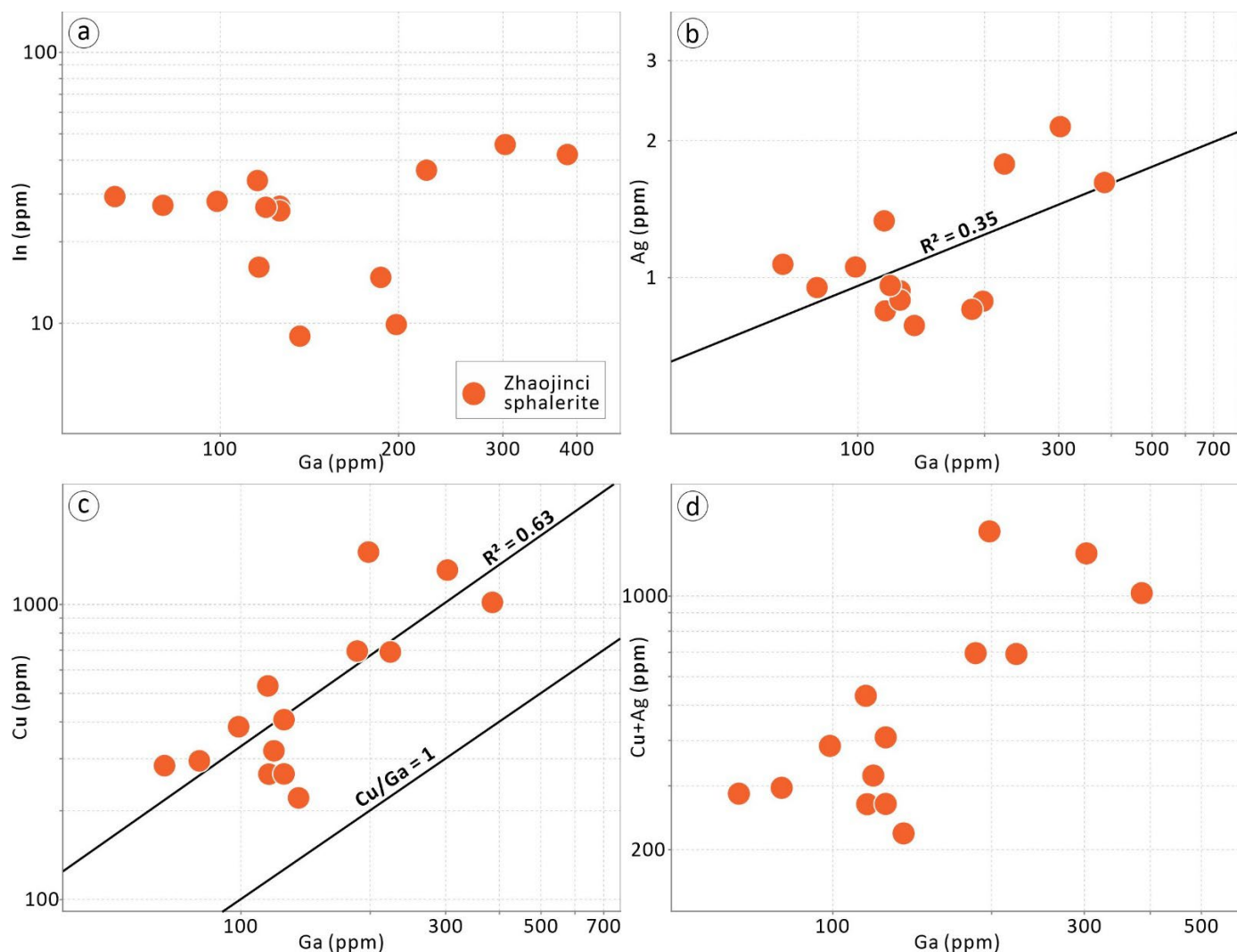


Figure 8. Binary plots of (a) Ga vs. In; (b) Ga vs. Ag; (c) Ga vs. Cu; (d) Ga vs. (Cu + Ag) for the Zhaojinci sphalerite.

5.3. Implications for Ore Origin and Mineral Prospecting

Trace elements in sphalerite serve as robust discriminators for deposit classification [59,89–91]. Typically, MVT sphalerite is characterized by elevated Ga, Ge, and Cd concentrations but depleted in Fe, Mn, In, Sn, and Co [77,89,92]. In contrast, magmatic-hydrothermal sphalerite is enriched in Fe, In, and Sn but depleted in Ga and Ge [77,89]. The Zhaojinci sphalerite exhibits a geochemical signature marked by high Ga, Ge, and Cd and low Mn, In, Sn, and Co (Figure 7; Table 3), consistent with MVT-like sphalerite. This classification is further supported by discriminant plots, including Fe–Mn (Figure 9a), In–Mn (Figure 9b), Ga–Mn (Figure 9c), and In–In–Ga (Figure 9d), where Zhaojinci sphalerite consistently plots within the MVT field, distinct compared to sphalerite from the magmatic-hydrothermal Dafang and Baoshan deposits (Figure 9).

While the Zhaojinci sphalerite exhibits this characteristic MVT signature (Figure 9), it is acknowledged that trace element contents alone are not uniquely diagnostic of a specific deposit type, as they primarily reflect the temperature and oxidation state of the ore-forming fluid [20,31]. However, the classification of the Zhaojinci mineralization as MVT-like

is strengthened by a confluence of geological evidence beyond sphalerite geochemistry: (1) Host rock and mineralization style: the mineralization is strictly stratabound within the Devonian Xikuangshan Formation limestone, a common host for MVT deposits [93,94]; (2) Structural control: ore localization is controlled by NE-trending syn-sedimentary normal faults (Figure 3), which provided excellent pathways for fluid migration, a typical structural setting for MVT districts [95]; (3) Hydrothermal alteration: pervasive dolomitization of the limestone host rock is observed (Figure 5a), a hallmark of MVT systems [93,96]. The dolomitized portions exhibit a crystalline texture distinct from the original micritic limestone; (4) Absence of magmatic-hydrothermal affinity: critically, there is no spatial or temporal association with Jurassic magmatic activity. The absence of high-temperature alteration minerals (e.g., garnet, pyroxene) and skarn assemblages rules out a proximal magmatic-hydrothermal origin; (5) Physicochemical conditions: the mineralization formed at low temperatures (118–138 °C) and under intermediate sulfur fugacity conditions ($\lg fS_2 = -22.9$ to -21.2), as recalculated based on the pyrite stability field [97], which aligns with the conditions of many MVT deposits [6,84,85,96]; and (6) Machine learning validation: An XGBoost machine learning classifier, trained on a global sphalerite dataset [98], assigned all 14 Zhaojinci sphalerite analyses (100%) to the MVT, providing independent, data-driven support for this classification. Although the mineralization at Zhaojinci shares key features with MVT-type deposits, an alternative distal magmatic-hydrothermal model must be considered, especially in light of the widespread local magmatism and the absence of known MVT examples in South Hunan [26]. In such a model, the mineralizing fluids could have been sourced from the unexposed Jurassic granitic intrusions in the region. These fluids might have migrated laterally along permeable horizons (e.g., the Devonian Xikuangshan Formation limestone) and/or fault systems (e.g., the NE-trending syn-sedimentary normal faults), depositing ore minerals at a considerable distance from the magmatic source under low-temperature and intermediate- fS_2 conditions. Nevertheless, the distinct MVT-like trace element signature of the sphalerite, the lack of a zoning pattern towards a magmatic center, and the absence of a connecting alteration halo collectively favor a non-magmatic, MVT-like model for its origin. Collectively, this multi-faceted evidence strongly supports an MVT-like affinity for the Zhaojinci mineralization, as defined by established criteria [93–96].

Consequently, the South Hunan district exhibits a bimodal Pb–Zn mineralization system, comprising both magmatic-hydrothermal type (as exemplified by the Dafang and Baoshan deposits) and MVT-like mineralization (represented by the Zhaojinci area). This metallogenic characteristic demonstrates spatial and genetic consistency with the mineralization patterns observed in the adjacent Hengyang Basin, where magmatic-hydrothermal systems are manifested in the Kangjiawan and Shuikoushan deposits [65,91], while MVT-like mineralization is prominently developed in the Qingshuitang deposit [84]. The identification of Ga-enriched MVT-like sphalerite at Zhaojinci provides concrete exploration vectors for mineral prospecting within the district. This discovery substantiates the critical role of low-temperature and intermediate fS_2 environments in Gallium enrichment and, more importantly, establishes a new exploration model by shifting the focus to include low-temperature, carbonate-hosted targets. Future exploration should prioritize areas with Devonian carbonate rocks (e.g., the Xikuangshan Formation), particularly where they are affected by syn-sedimentary faults and exhibit pervasive dolomitization. The distinct geochemical signature of sphalerite (Ga–Ge–Cd-rich and In–Sn–Mn-poor) serves as a direct fingerprint for identifying such mineralization, thereby providing a novel and effective paradigm for resource exploration in this metallogenically significant region.

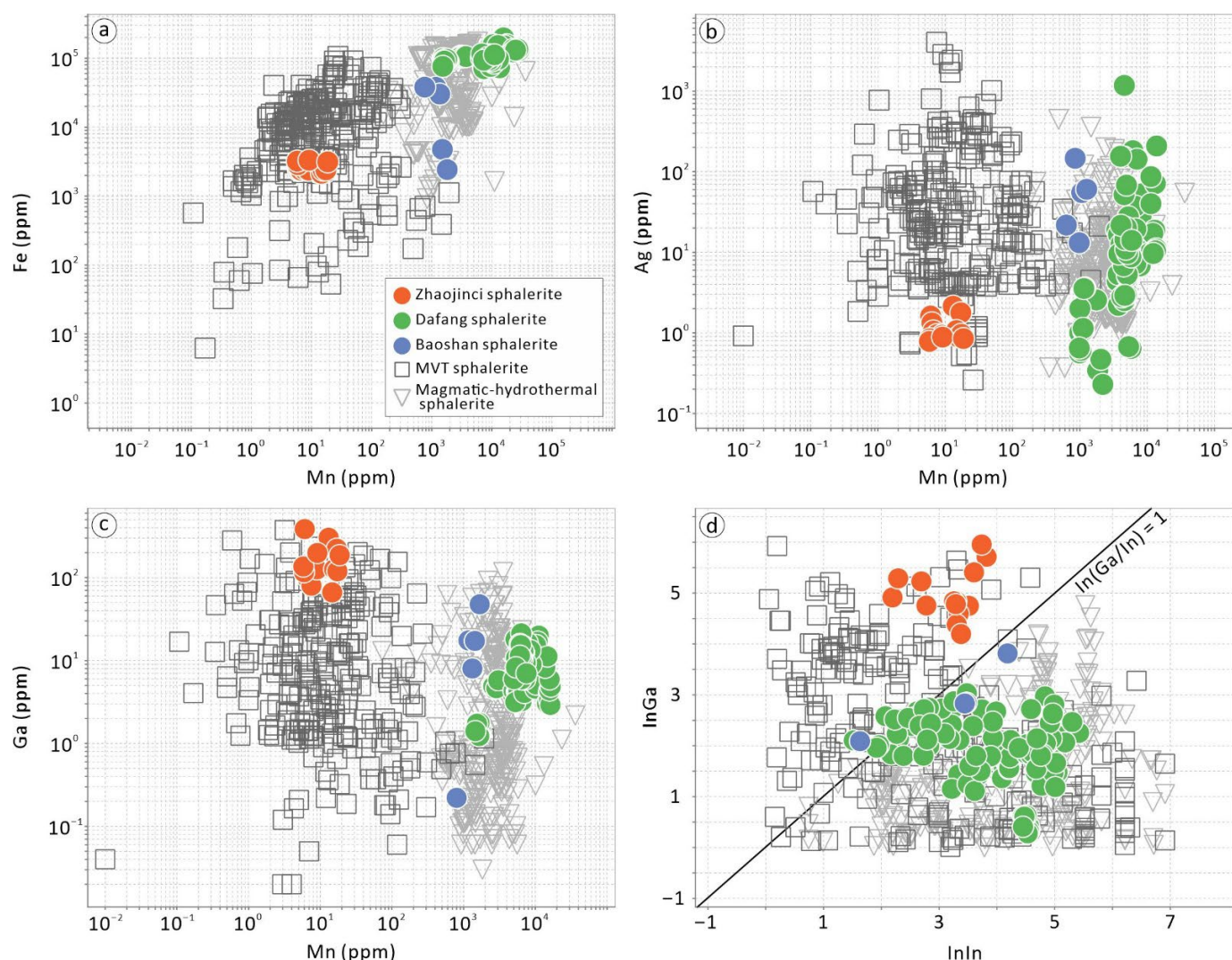


Figure 9. Binary plots of (a) Mn vs. Fe; (b) Mn vs. Ag; (c) Mn vs. Ga; (d) $\ln(\ln \text{ concentration})$ vs. $\ln(\ln \text{ Ga concentration})$ for the Zhaojinci sphalerite. The axes in (d) represent the natural logarithm (base e) of the element concentrations in parts per million (ppm). All Zhaojinci data are from LA-ICP-MS analyses. Data are compiled from published literature [6,15,21,29,30,59,71,84,91,99–107].

6. Conclusions

The South Hunan district, strategically located at the intersection of the Nanling Region and the Qin-Hang Metallogenic Belt, is renowned for its Jurassic magmatic-hydrothermal Pb–Zn deposits, typically characterized by Ga-depleted sphalerite. However, recent drilling at Zhaojinci has revealed Ga-enriched sphalerite (up to 385 ppm by LA-ICP-MS), adding complexity to regional metallogeny. The EPMA and LA-ICP-MS analyses demonstrate that Ga primarily substitutes for Zn^{2+} in the sphalerite lattice via coupled substitution with Cu^+ . Sphalerite geothermometer indicates low formation temperatures (118–138 °C), complemented by intermediate sulfur fugacity ($\lg fS_2 = -22.9$ to -21.2), which are critical for Ga enrichment. This evidence combined with pervasive dolomitization of Devonian limestone, mineralization controlled by syn-sedimentary normal faults, and the absence of magmatic or high-temperature hydrothermal signatures, supports an MVT-like affinity for the Zhaojinci mineralization. This study presents a multi-proxy case for classifying the Zhaojinci mineralization as an MVT-like deposit based on sphalerite geochemistry, physicochemical conditions, and geological setting. We acknowledge that more detailed studies, particularly fluid inclusion microthermometry and stable isotope (e.g., S, C, O) analyses,

would be highly valuable to further constrain the source and nature of the ore-forming fluids. These approaches represent important directions for future research to build upon this preliminary classification. This discovery not only elucidates the role of low-temperature and intermediate- fS_2 conditions in Ga enrichment but also highlights the potential for MVT-like Pb–Zn mineralization in South Hunan, guiding future prospecting towards carbonate sequences exhibiting the diagnostic Ga–Ge–Cd-enriched sphalerite signature.

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